



昆山杜克大学创新孵化器
DKU Innovation Incubator

Dii Midterm Reports

Project Code: IE169

Project Name: Radikal (previously Chinese Radical Lab)

Team Name: Wision

Team Logo: <https://duke.box.com/s/vdvsse0oxrqnwog7oa0v5o85pvi940mn>



Project Summary (50-150 words):

"Radikal" is an entrepreneurship project developing a gamified, AI-powered mobile application to address the significant challenge of memorizing Mandarin Chinese characters. Unlike rigid, curriculum-based platforms, it allows learners to import vocabulary from any source and utilizes interactive character construction, personalized memory associations, and AI-generated stories to create meaningful, emotional connections with the language.

Targeting tech-savvy young learners globally, the app employs a freemium model, monetizing through AI token purchases and institutional licenses. By transforming rote memorization into an engaging, creative activity, the project aims to make Chinese more accessible, support quality education (SDG 4), and foster greater cross-cultural understanding.

Team Member(s):

Name	Net ID	Class	Major	Role
Bilguun Dugarsuren	bd194	2028	Applied Mathematics and Computational Science in Computer Science track	Survey and logic flow designer, Developer, Team Treasurer
Nemuulen Togtbaatar	nt152	2027	Computation and Design in Computer Science track	Data collection & survey distributor, Developer, Team Leader, Communications



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1) Major achievements (500 words):

Over the past quarter, our team has made significant progress in transforming the Radikal (previously Chinese Radical Lab) project from an early concept into a functional, research-supported learning prototype with a clearer product direction. Our accomplishments span technical development, user research, community engagement, and internal team growth.

Our first major milestone was the successful development of a working web-based prototype. As it is easier to iterate after feedback, we decided to first develop a web version of our app. Although not yet fully refined, this prototype effectively demonstrates the core learning experience we envisioned: radical-based Chinese character learning, AI-driven story elements, and game features. We also tried to improve the user interface and overall visual design, transforming early wireframes into a smoother, more intuitive layout. Several interactive components, including basic navigation flow and gamification features, have been implemented, forming a foundation we will later transform into a full mobile application.

To evaluate whether our learning methodology resonates with real learners, we conducted our first public demonstration during the Chinese Teatime event that happened on the 5th of November, 2025, with over 30 participants, where we pitched our radical-based learning technique and early prototype to international students studying Chinese at various proficiency levels. This session served two key purposes: (1) testing the clarity, intuitiveness, and value of our teaching method, and (2) gathering immediate, qualitative feedback from target users in a natural learning environment. The insights we received helped validate the potential of combining radicals, stories, and visuals, and they highlighted areas where learners still required more guidance, interaction, or personalization within the app.

Following this initial demo, we launched a broader feedback survey targeted at international students across DKU. The survey collected data on students' learning habits, their perception of our prototype design, and their interest in using an app built around our method. Importantly, these results provided early evidence that our technique is promising and that learners are open to adopting story-based approaches to memorizing characters. The survey also helped us identify several pain points in the current design and informed the direction of our next development phase.

Another significant achievement lies in our team development. We participated in a team management workshop and mentoring sessions with entrepreneur and instructor Will Sung. These sessions gave us practical tools for effective communication, realistic goal-setting, and early-stage startup coordination. The mentorship also helped us refine our internal workflow,



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establish a clearer division of responsibilities, and incorporate structured decision-making methods. As a result, our team has become more collaborative, more focused, and more efficient in balancing creative experimentation with disciplined development.

Overall, this quarter allowed us to build a showcase prototype, validate our early ideas with real users, collect meaningful research data, and strengthen our team dynamics. These achievements together set a solid foundation for the next stage of development, where we will transition into mobile app implementation, improved AI integration, and deeper user testing.

2) How was your team's progress (highlight the one that matches)?

Ahead of Schedule **As Scheduled** Behind Schedule

3) What challenges do you have (300 words)?

Throughout the past quarter, our team encountered several challenges that shaped the trajectory of this project and influenced how we planned our development process.

One of the most significant challenges stemmed from our limited experience in full-scale app design and development. Although we initially intended to build a mobile application from the beginning, we quickly realized that our team lacked the technical depth, both in UI/UX design and mobile engineering. This led us to shift temporarily to a web-based prototype. Additionally, to support key features, such as personalized learning, character-of-the-day generation, and future GenAI story integration, we urgently needed a structured and comprehensive character dataset. Obtaining an open and reliable dataset and working on it proved more difficult than expected and became a bottleneck for several planned features. Moreover, due to a lack of experience, we are still hesitant about choosing the most effective platforms for hosting, AI integration, and such.

Another challenge emerged during our prototype feedback phase. Our first attempt at a feedback survey produced low-quality responses due to flaws in the website link, unclear instructions, and insufficient context for participants. This made the initial data almost unusable and forced us to rethink our research approach. We redesigned the survey, clarified our goals, and paired it with in-person demonstrations during the Chinese Teatime event. This improvement in methodology significantly increased the quality and depth of feedback, but the setback emphasized the importance of careful survey construction and user context.



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These challenges have been valuable learning experiences. They have guided us toward more realistic planning, stronger research practices, and clearer developmental milestones for the upcoming quarter.

4) Next half-year project plan:

Quarter 3 (Next 3 Months): Core App Foundation & Data Preparation

Main Goal: Build a stable foundation for the website & mobile app and prepare all the essential foundational data for learning.

1. Technical Development

- Improve the UI for a better user experience and understanding, and add a step-by-step tutorial for new users.
- Convert the improved Figma UI into actual React Native components using Expo + expo-router.
- Build the core navigation flow (Home → Character of the Day → Story → Creative Lab → Profile).
- Set up backend connection using Supabase.

2. Dataset & Content Preparation

- Finalize the HSK 1–4 character dataset (radicals, meanings, pinyin, examples).
- Structure the dataset into a clean JSON format for fast querying and future AI integration.
- Prepare a list of radicals, radical meanings, and radical-based hints.

3. Early User Testing & Feedback

- Identify weak points in UI, navigation, or explanation.
- Conduct small-scale usability tests with international students after improving the UI and backend to collect early feedback on visual clarity, flow, and ease of use.



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Quarter 4 (Following 3 Months): Core Features, AI Integration & Gamification

Main Goal: Implement interactive learning features and test AI-generated content.

1. Core Feature Implementation

- Develop Character of the Day module using patterns from the dataset + randomization logic.
- Implement Personalized Story Generation using OpenAI/DeepSeek prompts.
- Build the Creative Lab, enabling radical combination, guessing meaning, and basic scoring.
- Add the search input section (type or draw a character to explore).

2. Personalization & Testing

- Implement Profile Setup (age + interests → story topics).
- Begin testing personalized stories based on profile attributes.
- Conduct targeted feedback sessions with international Mandarin language learners at various institutions.

3. Gamification & Engagement

- Improve the rewards system logic: streaks, XP, and progress badges.
- Develop the leaderboard logic for friendly competition.
- Improve story flow and radical-based hints for faster learning.

4. Technical Optimization

- Improve API efficiency for mainland China users (caching, local endpoints).
- Collect performance metrics such as load time, story generation delay, and navigation smoothness.



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5) Financial status and next half-year budget plan:

Category	Item	Cost (CNY)	Notes
AI	ChatGPT Plus	140/month	Mainly for research
Graphics	Canva Pro	91/month	Design or download graphics to include in the UI
AI	Blackbox Pro	65/month	Mainly for technical support
Backend Hosting	Supabase Pro	150/ month	Our backend is already on Supabase
APIs	ChatGPT API (gpt-4o-mini / gpt-4o)	0.03–0.3/ 1K tokens	For the final product
APIs	DeepSeek API	0.006–0.02/ 1K tokens	Cheaper option for testing
Apple App Store	Apple Developer Program	720/year	Mandatory for publishing apps on iOS
Google Play Store	Google Play Developer Account	180/ one time	no annual renewal
Incentives	Final user test feedback	15/person	50 people's feedback will cost 750rmb, will be students from other institutes
Feedback	Visit other institutes(NYUSH, XJLTU, etc.) for feedback and experience sharing	500	Expected transportation and fees



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6) Supporting material:

Please attach any deliverables you have produced so far to demonstrate your project achievement. No limit on the format. (For example: Publication, certificate, etc.)

1. Chinese Teatime event, November 5th, 2025.



2. Prototype link: <https://dahlia-seven-32363242.figma.site>
3. Prototype feedback survey: https://duke.qualtrics.com/jfe/form/SV_9FziT3kvtJ3KdPE
4. Prototype feedback results report:
<https://duke.box.com/s/b9s7anhhe4h10zxt4b09uzepclmfuzj>

7) Supervisor comments:

The Wision Team has made steady progress towards the project goals. This midterm report is clear, and it shows promising preliminary outcomes. I hope the team can learn from the challenges and improve accordingly in the second half.

Supervisor signature:

Date: November 20th, 2025